



Adyant Agrawal

Postdoctoral researcher,
Institute for Computational
Physics, Stuttgart

Interests

- Fluid-Structure Interactions
- Interfacial effects
- Hydrodynamics
- Low-Reynolds number flows
- Porous Media Flow
- Multiscale Modelling
- Confined Electrolytes

Methods

- Molecular Dynamics
- Finite Element Methods
- Mesoscale hydrodynamics
- Free-energy methods
- Monte Carlo sampling
- Stochastic Dynamics
- Machine Learning
- Software Development

Tools

- GROMACS
- LAMMPS
- COMSOL
- Python
- MATLAB
- C++
- Git, Github, Gitlab

Education

Ph.D. in Mechanical Engineering

Dec 2018 - Sep 2022

School of Engineering and Material Science
Queen Mary University of London

Thesis title: Liquid exfoliation and rheology of 2D nanomaterials

Thesis supervisor: Prof. Dr. Lorenzo Botto

Bachelor of Technology in Engineering Physics

Jul 2013 - Nov 2017

Department of Physics
Indian Institute of Technology Delhi

Thesis title: Collective motion of self-propelled rods in quasi-2D geometry

Thesis supervisor: Prof. Dr. Sujin Babu

Research Experience

Postdoctoral Researcher

Jan 2023 - Present

Institute for Computational Physics, University of Stuttgart

Developed and applied atomistic, machine-learned, and multiscale simulation methods to investigate confined and interfacial fluid phenomena:

Doctoral Student

Dec 2018 - Sep 2022

Queen Mary University of London

Developed continuum hydrodynamics and atomistic frameworks to investigate fluid-structure interactions in confined and sheared flows:

Bachelor's Student

May 2016 - Nov 2017

Indian Institute of Technology Delhi

Developed computational models to investigate collective dynamics, hydrodynamic interactions, and transport phenomena in active and cosmological systems.

Publications

- A. Agrawal, C. Kamal, S. Gravelle, and L. Botto, Negative intrinsic viscosity in graphene nanoparticle suspensions induced by hydrodynamic slip, *ACS Nano* 19, 38350–38359 (2025). DOI: 10.1021/acsnano.5c10415.
- A. Agrawal, S. Gravelle, C. Holm, and A. Schlaich, Interfacial ion hydration and electrostatics govern salt precipitation and crystal morphology, *J. Chem. Phys.* 163 (2025). DOI: 10.1063/5.0299741.
- A. Agrawal, S. Gravelle, C. Kamal, and L. Botto, Viscous peeling of a nanosheet, *Soft Matter* 18, 3967–3980 (2022). DOI: 10.1039/D1SM01743H.
- A. Agrawal and S. B. Babu, Self-organization in a bimotility mixture of model microswimmers, *Phys. Rev. E* 97, 020401(R) (2018). DOI: 10.1103/PhysRevE.97.020401.
- A. Agrawal, Liquid exfoliation of graphite and rheology of graphene suspensions: Continuum and molecular simulations of fluid-structure interactions at the nanoscale, Ph.D. Thesis, Queen Mary University of London.
Repository: Queen Mary Research Online (QMRO)

Certifications

2018
Advanced Training in
Statistical Physics
Freie Universität Berlin

2022
Diploma of Researcher
Development
Queen Mary University of
London

Positions of Responsibility

2020 - 2021
Program Leader
Ekavya Underprivileged
Education Trust, India

2015 - 2016
Department Convener
Co-curricular and Academic
Interaction Council, IIT Delhi

2015 - 2016
Batch Representative
Board for Academic Programs,
IIT Delhi

2014 - 2015
Student Representative
Board for Student Welfare, IIT
Delhi

Contact

📍 Stuttgart

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Awards & Academic Distinctions

- 2025 Best Poster, Condensed Matter and Quantum Materials Conference, Bristol
Awarded for: "Negative intrinsic viscosity in graphene suspensions: insights from molecular dynamics and continuum theory."
- 2022 Queen Mary Doctoral Research Studentship in the School of Engineering and Materials Science
- 2017 Summer Research Fellowship, Indian Institute of Science Education and Research (IISER) Mohali, India
- 2015 Formula Student Design Challenge, Electronics Engineer — Team rank 9
- 2013 All India Percentile 99.5, IIT Joint Entrance Examination

Contributed Talks

- 2025 The role of hydration and electrostatics in salt crystal growth – 9th International Soft Matter Conference, Minoa Palace, Crete, Greece
- 2025 Salt precipitation in porous media: atomistic coupling of hydration and electrostatics – CECAM International Symposium 2025: Molecular Modelling in Soil Science – Achievements and Perspectives
- 2025 Molecular insights into irregular growth of salt crystals: the role of charge and water structure – Condensed Matter and Quantum Materials (CMQM) Conference 2025, University of Bristol, UK
- 2024 The interplay between hydration and growth of a salt crystal – Liquid Matter Conference 2024, Johannes Gutenberg University Mainz, Germany
- 2024 Role of hydration and electrostatics in salt precipitation – DPG Spring Meeting 2024, Berlin, Germany
- 2022 Viscous peeling of graphene sheets: towards producing graphene on the ton scale – European Solid Mechanics Conference 2022
- 2021 Flow-induced dispersion of 2D nanomaterials in sheared liquids – 4th Russell Binions Research PhD Symposium, Queen Mary University of London
- 2021 Viscous peeling of a graphene sheet – APS March Meeting 2021
- 2021 Can simulations help improve graphene production? – Industrial Engagement Event 2021, Queen Mary University of London
- 2018 Swimmers with different motility self-segregate – Computational Principles to Organise Complexity, Summer School, Nice, France